

Chapter 9: Heat

Textbook Exercise Solutions & Answers



1. Fill in the blanks:

(a) Maximum heat is absorbed by a coloured object.

black

(b) of heat does not require a medium.

Radiation

(c) Conduction of heat takes place through a substance.

solid

(d) The shining surface in a thermos flask decreases the outgoing heat by

reflection

(e) Cooking utensils are made from metals due to their property of

conduction

(f) The earth receives heat from the sun by

radiation

2. What will absorb heat?

Steel spoon, wooden board, glass vessel, iron griddle (lava), glass, wooden spoon, plastic plate, soil, water, wax.

What will absorb heat? Steel spoon, wooden board, glass vessel, iron griddle (lava), glass, wooden spoon, plastic plate, soil, water, wax.

Following will absorb heat fast: steel spoon, iron griddle, water, wax. Remaining objects also absorb heat but very slowly.

3. Write Answers to the following questions.

(a) How does a fever get lowered by putting a cold compress on the forehead of a patient?

The water in the cold compress will evaporate by the heat of fever thus taking some local heat of the body and temperature of the body is lowered, therefore the fever gets lowered by putting a cold compress on the forehead of a patient.

(b) Why are the houses in Rajasthan painted white?

Rajasthan being a desert, temperature is very high. So there is excess heat. White colour doesn't absorb the heat but reflects back, so houses will not become hot due to excess heat and inside of the house will be cool. So the houses in Rajasthan are painted white.

(c) What are the modes of heat transfer?

There are three modes of heat transfer:

1. conduction
2. convection
3. radiation.

(d) Explain which mode of heat transfer causes sea breezes and land breezes?

1. Unequal heating of air above the sea and land sets up convection currents which cause sea breezes during the day time.
2. The land near the sea is heated by sunlight to a higher temperature than the sea.
3. As the air above the land is heated, it becomes lighter and moves upwards, its place is then taken by cooler air moving from the sea towards the land.
4. Sea breezes are thus set up. At night, the land cools faster than water in the sea. Therefore air above sea water is hotter than air above the land.
5. So cooler air from the land moves towards the sea forming land breeze.

(e) Why is the outer coat of the penguins of Antarctica black?

The outer coat of the penguins of Antarctica is black because black colour absorbs heat so penguins get lots of heat due to this black colour and they can live easily in very, very cold Antarctica region.

(f) Why are heaters fitted near the floor and air conditioners near the ceiling of a room?

1. Due to air conditioners, air becomes cool so its density increases and it becomes heavy. So it goes down and warm air with less density comes up and takes the place of cold air. This cycle goes on and on and the whole room becomes cool.
2. If it is fitted near the floor then cool air of high density can't go up and upper air with low density cannot come down and cooling of the room does not take place properly so air conditioners are fitted near the ceiling of a room.
3. Heaters are fitted near the floor. Due to heating, air becomes hot which is lighter than cold air. So it goes up and upper air with high density comes down.
4. This cycle goes on and on and the full room becomes warm. So heaters are fitted near the floor.

4. Give scientific reasons.

(a) An ordinary glass bottle cracks when boiling water is poured into it, but a borosil glass bottle does not.

1. An ordinary glass bottle cracks when we pour boiling water into it because there is a lot of temperature difference between the inner and outer parts of the glass.
2. The inner part gets hot and expands while the outer part remains cold since glass is a bad conductor of heat.
3. This unequal expansion causes the crack of glass bottle.
4. But the borosil glass bottle has a very high melting point so the temperature difference is very less, hence it doesn't crack.

(b) The telephone wires which sag in summer become straight in winter.

1. Telephone wires are made up of copper metal. Metals expand on heating.
2. In summer the copper wire expands, thus the length of the wire increases between the two poles and instead of remaining straight it sags.
3. In winter due to cold, it contracts and comes back to the original position, so it becomes straight in winter.

(c) Dewdrops form on the grass in winter.

1. During the winter, the temperature of air is low, its capacity to hold the vapour is less.
2. At such times the excess water vapour is transformed into water droplets due to cold. And so the dew drops form on the grass in winter.

(d) In winter, why does an iron pillar feel colder than a wooden pole.

1. Iron is a metal and metals are very good conductors of heat so it transfers heat at a much faster rate than wood.
2. That means, when we touch them, metals conducts away energy in the form of heat emitted by our hand and temperature of our hand lowers down and we get a cold' sensation.
3. This is what makes the metal appear colder than the wood.
4. The wood is taking away heat at a much slower rate.